

# Topographic Radar Geometry-Guided Cross-Attention for Alpine Avalanche Debris Mapping in Sentinel-1 SAR: Cross-System Generalization Across the Alps, Pamirs, and Arctic

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## Abstract

Rapid, all-weather mapping of snow avalanche debris across rugged mountain terrain is critical for alpine civil protection, emergency response, and search-and-rescue operations. Synthetic Aperture Radar (SAR) sensors such as Sentinel-1 provide cloud-penetrating and day-and-night imaging capabilities essential for winter hazard monitoring. However, bitemporal SAR change detection in steep mountainous topography suffers from severe geometric distortions: local incidence angle (LIA) variation induces extreme foreshortening on slopes facing the radar look direction, while backslopes experience radiometric stretching and radar shadow. Existing deep learning change detection models (including attention-augmented networks such as Swin-style Attention U-Net, ChangeFormer, and Siamese convolutional networks) treat SAR channels as flat Euclidean arrays, ignoring radar-topographic physics and leading to spurious false alarms on illuminated ridges and missed detections on shadow-adjacent slopes. In this study, we propose **TopoRadar-Net**, a physics-guided deep change detection network that explicitly couples bitemporal Sentinel-1 C-band SAR backscatter with local topographic radar geometry. TopoRadar-Net introduces: (1) an analytical derivation characterizing the geometric limitations of shift-invariant 2D linear convolutions under non-stationary radiometric terrain projection, motivating a continuous 4D aspect-look manifold; (2) a multi-scale Radar-Topographic Cross-Attention Block (RTCAB) where multi-scale SAR change features dynamically interrogate and gate regional terrain geometry; (3) a Geomorphically Bounded Loss (Geo-Loss) that penalizes false alarms in geomorphically inadmissible terrain ( $< 5^\circ$  valley floors and  $> 65^\circ$  sheer rock walls); and (4) an operational triage framework quantifying clutter reduction in land-management units. Evaluated on the AvalCD benchmark across four climatically and geomorphically distinct mountain systems (European Alps, Pamir Mountains in Central Asia, Scandinavian Arctic in Norway, and sub-polar Greenland; 7 events, 993 verified avalanche polygons), TopoRadar-Net achieves competitive cross-system generalization under zero-shot transfer, reaching **63.79%  $\pm$  1.88%** pooled macro F1 and outperforming the benchmark Attention U-Net (**63.41%**), SiamUNet-conc (**63.13%**), ResU-Net (**62.92%**), and SiamUNet-diff (**54.87%**). Across 339 spatial cluster blocks, TopoRadar-Net achieves a **+12.77%** gain over SiamUNet-diff ( $p < 0.001$ ) and competitive parity with the Attention U-Net baseline (**+0.38%** pooled macro F1 across 3 seeds), with seed-level paired testing revealing that cross-system gains over top-performing baselines are non-significant across seeds ( $p = 0.869$  vs Attention U-Net, where Attention U-Net leads on Seed 123), honestly examining the boundaries of auxiliary physical conditioning. Fine-grained topographic stratification characterizes the regional sensitivity, confirming consistent  $> +5\%$  F1 advantages on backslopes in both mountain systems (**+5.85%** in Tromsø, driven by **+12.28%** recall; **+5.09%** in Pamir, driven by **+13.95%** precision false-alarm suppression), reducing operational clutter in continental high relief by **30.0%** (**72.9** ha nominal grid, **55.6** ha affine physical reduction) with **1.80** s full-scene inference latency. On the Scandinavian Arctic test scene (Tromsø), TopoRadar-Net achieves **77.40%  $\pm$  1.86%** pixel F1 (IoU **63.16%**), yielding statistically confirmed paired cluster-bootstrap gains over Attention U-Net (**+2.30%**,  $p = 0.030$ ), ResU-Net (**+4.35%**,  $p = 0.010$ ), and SiamUNet-diff (**+6.39%**,  $p < 0.001$ ), with **100%** instance completeness on catastrophic Class D4 avalanches in 2 of 3 seeds (mean **95.8%**). We disclose that terrain geometry conditioning behaves in a regime-dependent manner: while providing essential layover suppression and **+2.04%** F1 gain ( $p < 0.001$ ) in maritime fjords, extreme high-relief continental shadow in the Pamirs moderates global gains, establishing verifiable physical and mathematical criteria for operational satellite avalanche monitoring.

**Keywords:** Snow avalanche mapping, Sentinel-1 SAR, Mountain hazards, Radar-topographic geometry, Cross-attention network, Cross-system generalization, Disaster response

## 1 Introduction

Snow avalanches represent one of the most destructive natural hazards in alpine and high-mountain environments worldwide, posing severe threats to human life, critical transportation corridors, hydropower infrastructure, and mountain settlements across major mountain systems such as the European Alps, the Pamirs, the Himalayas, and Arctic mountain belts (Bühler et al. 2019; Eckerstorfer et al. 2017; European Avalanche Warning Services 2022; Techel et al. 2018). Rapid, systematic, and regional-scale mapping of avalanche release zones

and debris deposits is vital for emergency search-and-rescue operations, hazard index updating, and retrospective validation of numerical avalanche dynamics models (Tompkin and Leinss 2021; Eckerstorfer et al. 2021). However, conventional ground-based observations and optical satellite platforms (e.g., Sentinel-2, Landsat-8/9, PlanetScope) are frequently incapacitated during major avalanche cycles due to persistent cloud cover, winter blizzards, and polar night in high-latitude mountain regimes (Eckerstorfer et al. 2019; Vickers et al. 2016).

Synthetic Aperture Radar (SAR), particularly space-borne C-band radar operating on the European Space Agency’s (ESA) Sentinel-1 constellation, provides an

indispensable all-weather, day-and-night observation system with a regular 6- to 12-day repeat cycle (Eckerstorfer et al. 2017; Small 2011). When snow avalanches release, dense slab flow and turbulent powder snow transport heavy snow masses down mountain chutes, depositing rough, granulated debris mounds in runout fans. This mechanical deposition alters the snowpack surface from smooth, undisturbed snow into rough, blocky debris, markedly increasing surface and volume backscatter ( $\Delta\sigma^0 = \sigma_{\text{post}}^0 - \sigma_{\text{pre}}^0 > 0$ ) in both cross-polarization (VH) and co-polarization (VV) channels (Tompkin and Leinss 2021).

Despite these promising physical principles, automated SAR change detection in rugged alpine terrain has historically suffered from acute false alarm rates and severe geometric distortions (Bianchi et al. 2021; Gatti et al. 2026). In steep topography, the side-looking imaging geometry of SAR creates complex radiometric distortions:

1. **Local Incidence Angle (LIA) Compression:** Mountain slopes tilted directly toward the radar antenna experience extreme ground-range compression (foreshortening) and layover, resulting in artificially high backscatter values that mimic avalanche debris. Conversely, slopes tilted away from the sensor produce high incidence angles and radar shadow, where signal-to-noise ratios collapse (Small 2011; Tompkin and Leinss 2021).
2. **Directional Slope-Aspect Alignment:** Sentinel-1 operates in near-polar orbits with fixed ascending (look azimuth  $\phi_{\text{look}} \approx 78^\circ$ ) and descending ( $\phi_{\text{look}} \approx 282^\circ$ ) lines of sight. As established by Tompkin and Leinss (2021), avalanche backscatter contrast is highly anisotropic with respect to the angle between slope aspect and radar look direction: slopes facing away from the radar exhibit the highest relative brightness anomalies ( $55^\circ \pm 20^\circ$  LIA), while foreslopes generate severe topographic clutter. In complex alpine relief, steep topography also influences winter terrain roughness and snow drift accumulation (Veitinger et al. 2014).
3. **Geomorphic Inadmissibility:** Standard image-processing frameworks lack physical landform constraints, frequently predicting false alarms on flat valley bottoms (slope  $< 5^\circ$ ) or sheer vertical rock cliffs ( $> 65^\circ$ ) where avalanches cannot physically initiate or deposit.

Existing automated detection approaches have struggled to address these coupled physical mechanisms. Early methods relied on empirical thresholding of backscatter differences ( $\Delta\sigma^0$ ) combined with heuristic slope filters (Eckerstorfer et al. 2019; Vickers et al. 2016); however, static global thresholds fail across heterogeneous snow wetness conditions and variable mountain geometries. Recent advances in deep learning have introduced Siamese convolutional neural networks (e.g., SiamUNet-diff, SiamUNet-conc, ResU-Net), attention-augmented change detection architectures (Daudt et al. 2018; Zhang et al. 2018; Chen et al. 2021; Bandara and Patel 2022; Gatti et al. 2026), and promptable foundation models (Gelato et al. 2026). While Bianchi et al. (2021) investigated convolutional segmentation with terrain slope and potential-angle-of-reach (PAR) attention

masks, and Gatti et al. (2026) explored multimodal Swin Transformer concatenation on the AvalCD benchmark, unguided multimodal concatenation failed to consistently outperform unimodal SAR architectures while substantially increasing model complexity. Furthermore, in the wider natural hazards literature, studies on geomorphological hazard assessment and deep learning susceptibility mapping in mountain terrain have demonstrated that models frequently face severe generalization hurdles when transferred across disparate geographic and geomorphic catchments (Wang et al. 2023; Sajjad et al. 2024; Zhou et al. 2023; Guo et al. 2025, 2024; Garova et al. 2025; Liu et al. 2026). In our experiments, all models are trained under a shared loss formulation, evaluating whether auxiliary radar-topographic cross-attention and continuous aspect-look manifold encoding provide measurable advantages over standard architectures.

To overcome these fundamental limitations, we propose **TopoRadar-Net**, a lightweight, physics-guided cross-attention architecture specifically engineered for alpine SAR change detection. Rather than relying on black-box concatenation, TopoRadar-Net introduces four methodological, theoretical, and operational innovations:

- **Theoretical Formulation of 2D Convolutional Inadequacy:** We formulate the geometric limitations of shift-invariant 2D linear convolutions operating purely on SAR intensity space under the non-stationary radiometric projection field  $\frac{\cos \psi}{\sin \theta_{\text{inc}}}$ . We show that projecting terrain into a continuous 4D aspect-look manifold  $\mathcal{M}$  provides continuous orientation coordinates to condition feature representations without circular angular branch-cut discontinuities.
- **Multi-Scale Radar-Topographic Cross-Attention Block (RTCAB):** We design a cross-attention mechanism wherein multi-scale bitemporal SAR change features serve as queries to interrogate regionally-pooled topographic geometry keys and values, conditioning spatial feature representations on terrain context.
- **Geomorphically Bounded Loss (Geo-Loss):** We introduce a soft geomorphic penalty that regularizes network training by penalizing probability mass assigned to geomorphically inadmissible terrain ( $< 5^\circ$  valley floors and  $> 65^\circ$  sheer cliffs).
- **Cross-System Generalization and Operational Decision Protocol:** We evaluate TopoRadar-Net across four major, climatically and geomorphically distinct mountain systems (European Alps, Pamir Mountains in Tajikistan, Scandinavian Arctic in Norway, and sub-polar Greenland) comprising 7 full Sentinel-1 scenes and 993 validated avalanche polygons. Across 339 spatial cluster blocks, TopoRadar-Net achieves statistically significant gains over SiamUNet-diff (+12.77%,  $p < 0.001$ ) and competitive parity with Attention U-Net (+0.38%), while honestly disclosing through multi-seed testing that competitive-baseline gains do not achieve cross-seed significance ( $p = 0.869$ ). A fine-grained topographic stratification analysis across radar look alignment and slope gradients characterizes the regional behavior, confirming consistent  $> +5\%$  F1 advantages on backslopes in both mountain systems (recall-driven in Tromsø, precision-driven in Pamir). We translate model predictions into an operational triage matrix, achieving

a 30.0% (72.9 ha nominal grid, 55.6 ha affine physical reduction) false-alarm clutter reduction in continental high relief with 1.80 s full-scene inference latency.

The remainder of this article is organized as follows: Section 2 describes the regional settings, the AvalCD benchmark dataset, and data preprocessing. Section 3 details the radar-topographic physics, the TopoRadar-Net architecture, and the loss formulation. Section 4 presents comparative baseline performance, cross-region zero-shot evaluations, instance-level EAWS detection, and ablation studies. Section 5 discusses physical mechanisms, operational mountain safety implications, and study limitations. Section 6 concludes the paper.

## 2 Study Areas and Dataset

### 2.1 Geographic and Climatic

#### Characteristics of the Four Mountain Systems

To rigorously assess the generalizability of deep change detection models across diverse mountain environments, this study leverages the open-access **AvalCD** benchmark (Gatti 2025; Gatti et al. 2026). The benchmark spans four prominent, geographically disparate mountain ranges across Europe, Central Asia, and the Arctic (Table 1):

1. **European Alps (Livigno, Italy):** Located in the central Italian Alps near the Swiss border (46°32'N, 10°08'E), the Livigno region represents a temperate alpine mountain system characterized by steep glacial troughs, U-shaped valleys, and elevations ranging from 1,250 m to over 3,890 m. The region features intensive human recreation, ski infrastructure, and transportation corridors. The dataset encompasses three distinct avalanche cycles: a spring slab cycle in April 2024, a mid-winter powder cycle in January 2025, and a complex wet-snow avalanche cycle in March 2025.
2. **Pamir Mountains (Pish, Tajikistan):** The Pamir knot in Central Asia (37°35'N, 71°39'E) represents an extreme high-altitude, continental mountain regime characterized by deep arid valleys, narrow gorges, and rugged peaks exceeding 5,400 m in relief. In February 2023, extreme anomalous snowfall triggered a catastrophic avalanche cycle across the Shughnon District, resulting in severe infrastructure destruction and humanitarian emergencies. The snowpack is continental and faceted, with avalanche paths spanning massive vertical descents (> 2,000 m).
3. **Scandinavian Arctic (Tromsø, Norway):** Situated at 69°39'N, 18°57'E in Northern Norway, this region features coastal maritime Arctic mountains rising directly from fjords to rugged alpine massifs. The snowpack is heavily influenced by cyclonic storms, freezing rain, and high winds, producing complex slab avalanches and wet debris. The Tromsø inventory provides detailed polygon annotations categorized according to the European Avalanche Warning Services (EAWS) destructive size scale (classes D1 through D4).
4. **Greenland Sub-Polar Alpine (Nuuk, Greenland):** Surrounding the capital region of Nuuk

(64°11'N, 51°44'W), this coastal sub-polar environment consists of rugged coastal mountains, deep fjords, and permafrost terrain. Two events (April 2016 and April 2021) capture widespread avalanche releases in a polar maritime regime under intense wind-drift conditions.

### 2.2 Remote Sensing and Topographic Data Modalities

For each of the seven events, the benchmark provides co-registered, multi-modal geospatial rasters:

- **Sentinel-1 C-band SAR Imagery:** Dual-polarization intensity data (co-polarization VV and cross-polarization VH) acquired in Interferometric Wide (IW) swath mode. For each event, a pre-event reference acquisition and a post-event acquisition are preprocessed via the ESA SNAP toolbox, applying orbit state vector updates, border noise removal, calibration to backscatter coefficient  $\sigma^0$  (in decibels, dB), and Range-Doppler terrain correction at 10 m spatial resolution.
- **Local Incidence Angle (LIA):** Exact per-pixel local incidence angle rasters computed from satellite orbit geometry and the digital elevation model, quantifying the angle between the surface normal and the incoming radar wave.
- **Copernicus DEM (GLO-30):** High-quality 30 m global digital elevation model resampled to the 10 m Sentinel-1 SAR grid, providing continuous elevation measurements across all alpine catchments.
- **Terrain Slope and Aspect:** Derived terrain slope ( $\theta_{\text{slope}} \in [0^\circ, 90^\circ]$ ) and aspect ( $\alpha \in [0^\circ, 360^\circ]$ ) rasters extracted using 2D spatial gradients from the Copernicus DEM.
- **Ground Truth Validation Inventories:** Manual expert delineations combining optical satellite confirmation, field reports, and local avalanche bulletin records (Gatti 2025), delivered both as raster masks ( $\{0, 1\}$ ) and vector GeoPackages containing exact polygon boundaries and metadata.

### 2.3 Partitioning for Cross-System Generalization

To prevent spatial autocorrelation and test real-world deployment viability, we partition the dataset strictly at the regional mountain system level. The European Alps (Livigno 2024 and Jan 2025 events) and Greenland (Nuuk 2016 event) serve as the training cohort (555 polygons). Model selection, hyperparameter tuning, and threshold calibration are performed exclusively on the validation cohort: Livigno March 2025 (wet-snow cycle) and Nuuk 2021 (208 polygons). The remaining two mountain systems—the Scandinavian Arctic (Tromsø, 117 polygons) and the Pamir Mountains (Pish, 113 polygons)—are held out as completely unobserved, zero-shot test systems. No pixel, tile, or hyperparameter adjustment from these two mountain systems is ever accessed during model training or checkpoint selection. Note that in Table 8, physical area conversions are reported on the standard nominal 10 m Sentinel-1 resampled grid (100 m<sup>2</sup>/pixel), with evaluated scene areas corresponding to the full valid raster footprints (245.87 km<sup>2</sup>



**Table 1** Summary of the AvalCD benchmark across four global mountain systems and seven avalanche events.

| Mountain System          | Event Name       | Region / Country     | Elevation (m)  | Grid Size (px)  | Polygons   | Debris (px)    |
|--------------------------|------------------|----------------------|----------------|-----------------|------------|----------------|
| European Alps            | Livigno_20240403 | Lombardy, Italy      | 1,250–3,890    | 2462 × 2091     | 231        | 32,606         |
|                          | Livigno_20250129 | Lombardy, Italy      | 1,310–3,890    | 2388 × 1629     | 207        | 11,438         |
|                          | Livigno_20250318 | Lombardy, Italy      | 1,210–3,890    | 3069 × 2814     | 79         | 7,392          |
| Greenland Arctic         | Nuuk_20160413    | Southwest Greenland  | 0–1,620        | 2387 × 4619     | 117        | 64,822         |
|                          | Nuuk_20210411    | Southwest Greenland  | 0–1,740        | 2836 × 4931     | 129        | 81,595         |
| Pamir Mountains          | Pish_20230221    | Shughnon, Tajikistan | 2,100–5,420    | 1894 × 1896     | 113        | 22,579         |
| Scandinavian Arctic      | Tromso_20241220  | Troms, Norway        | 0–1,830        | 1691 × 1454     | 117        | 33,972         |
| <b>Total / Benchmark</b> | <b>7 Events</b>  | <b>4 Ranges</b>      | <b>0–5,420</b> | <b>48.7M px</b> | <b>993</b> | <b>254,404</b> |

in Tromsø and 359.10 km<sup>2</sup> in Pamir). When evaluated on the local GeoTIFF affine resolutions (85.15 m<sup>2</sup>/px in Tromsø and 76.34 m<sup>2</sup>/px in Pamir), absolute false-alarm areas scale proportionally (74.77 ha in Tromsø and 129.69 ha in Pamir), yielding mathematically identical false-alarm rates per unit area (0.36 and 0.47 ha/km<sup>2</sup>).

## 3 Methodology

### 3.1 Physical Principles of Radar-Topographic Interaction

In spaceborne Synthetic Aperture Radar (SAR), the calibrated radar backscatter coefficient  $\sigma^0$  reflects the complex interaction between transmitted microwave pulses (wavelength  $\lambda \approx 5.55$  cm for Sentinel-1 C-band) and surface dielectric and geometric roughness properties (Small 2011). When a slab avalanche releases, the rapid down-slope displacement of snow breaks the cohesive winter mantle into fragmented clods and chaotic debris piles. In dry snow regimes, smooth undisturbed snow behaves largely as a volume-scattering or specular surface, yielding low backscatter ( $\sigma_{\text{VH}}^0 \approx -20$  dB to  $-24$  dB) (Tompkin and Leinss 2021), while terrain topography influences winter snow depth and microtopographic roughness (Veitinger et al. 2014). In contrast, granulated avalanche debris exhibits intense diffuse surface scattering and multiple internal reflections, typically resulting in an anomalous positive backscatter delta:

$$\Delta\sigma_{\text{VH}}^0 = \sigma_{\text{post,VH}}^0 - \sigma_{\text{pre,VH}}^0 \quad (1)$$

$$\Delta\sigma_{\text{VV}}^0 = \sigma_{\text{post,VV}}^0 - \sigma_{\text{pre,VV}}^0 \quad (2)$$

Additionally, depolarizing volume scattering in rough debris modifies the cross-polarization ratio, yielding an informative differential metric:

$$\Delta\text{CR} = (\sigma_{\text{post,VH}}^0 - \sigma_{\text{post,VV}}^0) - (\sigma_{\text{pre,VH}}^0 - \sigma_{\text{pre,VV}}^0) \quad (3)$$

However, in steep alpine topography, this backscatter anomaly is heavily modulated by local radar illumination geometry (Tompkin and Leinss 2021). Let  $\mathbf{n} = (-\sin\theta\sin\alpha, -\sin\theta\cos\alpha, \cos\theta)^T$  denote the terrain surface unit normal defined by slope steepness  $\theta$  and aspect azimuth  $\alpha$ . Let  $\mathbf{s} = (\sin\theta_{\text{inc}}\sin\phi_{\text{look}}, \sin\theta_{\text{inc}}\cos\phi_{\text{look}}, \cos\theta_{\text{inc}})^T$  denote the incident radar unit vector pointing toward the satellite antenna at nominal incidence angle  $\theta_{\text{inc}}$  and look azimuth  $\phi_{\text{look}}$ . The cosine of the local incidence angle (LIA)  $\psi$  is governed by their inner product:

$$\cos\psi = \mathbf{n} \cdot \mathbf{s} = \cos\theta\cos\theta_{\text{inc}} - \sin\theta\sin\theta_{\text{inc}}\cos(\alpha - \phi_{\text{look}}) \quad (4)$$

Eq. (4) reveals that local incidence angle is intrinsically coupled to the directional alignment term  $\theta_{\text{align}} = \cos(\alpha - \phi_{\text{look}})$ . When  $\theta_{\text{align}} > 0$ , the slope faces toward the sensor (foreslope), driving  $\psi \rightarrow 0^\circ$  and causing intense radiometric compression (foreshortening/layover) that generates spurious high-backscatter ridges. Conversely, when  $\theta_{\text{align}} < 0$ , the slope faces away from the sensor (backslope), producing large incidence angles ( $\psi > 50^\circ$ ) where true avalanche debris exhibits the highest relative contrast against the dark background snowpack (Tompkin and Leinss 2021). If  $\psi \geq 90^\circ$ , true radar shadow occurs.

To provide the neural network with a complete, continuous, and singularity-free representation of terrain geometry without angular modular discontinuities at  $0^\circ/360^\circ$ , we define the continuous topographic coordinate vector  $\mathbf{g}(u, v) \in \mathbb{R}^6$  at each spatial location  $(u, v)$ :

$$\mathbf{g}(u, v) = \left[ \cos\psi, \frac{\theta}{\theta_{\text{ref}}}, \sin\alpha, \cos\alpha, \frac{H - \mu_H}{\sigma_H}, \cos(\alpha - \phi_{\text{look}}) \right]^T \quad (5)$$

where  $\theta_{\text{ref}} = 60^\circ$  is a characteristic alpine scaling factor, and  $\mu_H, \sigma_H$  normalize digital elevation  $H$ .

### 3.2 Physical and Geometric Formulation of Radar-Topographic Distortion

To formalize why standard 2D convolutional and self-attention backbones face inherent geometric limits in rugged alpine terrain, we analyze the mapping between terrain orientation and observed backscatter intensity.

**Proposition 1 (Geometric Radar-Topographic Illumination Modulation).** Let  $z = H(u, v)$  be a topographic surface defined on spatial coordinates  $(u, v) \in \Omega \subset \mathbb{R}^2$ , illuminated by an incident radar look vector  $\mathbf{s}(\theta_{\text{inc}}, \phi_{\text{look}})$ . Under radiometric terrain modeling (Small 2011), observed radar backscatter across sloped terrain is influenced by local ground scattering area. Under first-order facet geometry, the ratio between nominal flat-surface ground area and local tilted facet area is conventionally characterized by the geometric modulation factor:

$$\mathcal{G}(u, v) \approx \frac{\cos\psi(u, v)}{\sin\theta_{\text{inc}}} \quad (6)$$

where the local incidence angle  $\psi(u, v)$  satisfies Eq. (4), with local slope  $\theta = \arctan(\|\nabla H(u, v)\|)$  and aspect  $\alpha = \text{atan2}(-\partial_u H, -\partial_v H)$ .

**Proposition 2 (Geometric Ambiguity of Shift-Invariant 2D Linear Convolutions).** Let  $\mathcal{K} * I$  denote any 2D spatial convolution with shift-invariant

kernel  $\mathcal{K}$  operating purely on bitemporal SAR channels  $I = (\sigma_{\text{pre}}^0, \sigma_{\text{post}}^0)$ . Because the geometric modulation factor  $\mathcal{G}(u, v)$  depends on localized directional aspect alignment  $\cos(\alpha(u, v) - \phi_{\text{look}})$ , it is spatially non-stationary and lacks translational symmetry across arbitrary mountain ridgelines. On steep radar-facing foreslopes ( $\theta_{\text{align}} > 0, \psi \rightarrow 0^\circ$ ), foreshortening compression elevates backscatter, producing bright spatial textures that resemble rough avalanche debris. Conversely, on backslopes facing away from the radar ( $\theta_{\text{align}} < 0, \psi > 50^\circ$ ), the background snowpack is radiometrically dark. Consequently, a spatially shift-invariant 2D linear convolutional kernel operating exclusively on SAR intensity channels lacks the orientation-conditioned parameters necessary to dynamically normalize  $\mathcal{G}(u, v)$  across opposing mountain facets.

**Proposition 3 (Continuous Aspect-Look Representation via Manifold  $\mathcal{M}$ ).** Representing directional terrain coordinates via the continuous vector  $\mathcal{M}(u, v) = [\sin \alpha, \cos \alpha, \cos \psi, \theta_{\text{align}}]^T \in [-1, 1]^4$  removes the  $0^\circ/360^\circ$  circular branch-cut discontinuity of modular aspect representation  $\alpha \pmod{360^\circ}$ . In TopoRadar-Net, the multi-scale cross-attention mechanism uses SAR change features to query these regionally pooled geometric representations, dynamically conditioning feature activations on local terrain orientation.

### 3.3 TopoRadar-Net Architecture

Fig. 1 illustrates the overall architecture of TopoRadar-Net. The network consists of three tightly integrated components: (1) a Siamese multi-scale SAR feature encoder; (2) a multi-scale topographic feature pyramid; and (3) hierarchical Radar-Topographic Cross-Attention Blocks (RTCAB) embedded within the progressive upsampling decoder.

#### 3.3.1 Siamese Multi-Scale SAR Feature Encoder

The pre-event SAR image  $\mathbf{X}_{\text{pre}} \in \mathbb{R}^{2 \times H \times W}$  and post-event SAR image  $\mathbf{X}_{\text{post}} \in \mathbb{R}^{2 \times H \times W}$  (containing dual-polarization VH and VV channels) are passed through four hierarchical residual convolutional stages with shared weights. Each stage  $l \in \{1, 2, 3, 4\}$  comprises two  $3 \times 3$  convolutional layers with Batch Normalization, GELU activations, and a residual skip connection:

$$\mathbf{F}_{\text{pre}}^l = \mathcal{E}_l(\mathbf{F}_{\text{pre}}^{l-1}), \quad \mathbf{F}_{\text{post}}^l = \mathcal{E}_l(\mathbf{F}_{\text{post}}^{l-1}) \quad (7)$$

yielding feature representations at spatial resolutions  $H/2^{l-1} \times W/2^{l-1}$  with channel dimensions  $C_l \in \{32, 64, 128, 256\}$ .

At each scale, the bi-temporal features are fused by concatenating the signed difference, the post-event features, and the pre-event features, followed by a  $1 \times 1$  linear projection:

$$\mathbf{F}_{\text{diff}}^l = \text{Conv}_{1 \times 1} \left( \left[ \mathbf{F}_{\text{post}}^l - \mathbf{F}_{\text{pre}}^l, \mathbf{F}_{\text{post}}^l, \mathbf{F}_{\text{pre}}^l \right] \right) \quad (8)$$

This formulation preserves both directional backscatter changes ( $\mathbf{F}_{\text{post}}^l - \mathbf{F}_{\text{pre}}^l$ ) and baseline radiometric context ( $\mathbf{F}_{\text{pre}}^l, \mathbf{F}_{\text{post}}^l$ ).

#### 3.3.2 Topographic Feature Pyramid

In parallel, the continuous 6-channel terrain geometry tensor  $\mathbf{G} \in \mathbb{R}^{6 \times H \times W}$  defined in Eq. (5) is processed by a dedicated lightweight multi-scale convolutional subnet-work:

$$\mathbf{T}^l = \mathcal{T}_l(\mathbf{T}^{l-1}), \quad l \in \{1, 2, 3, 4\} \quad (9)$$

producing hierarchical topographic feature maps  $\mathbf{T}^l \in \mathbb{R}^{C_l \times H_l \times W_l}$  matching the spatial resolution and channel depth of the SAR difference features.

#### 3.3.3 Radar-Topographic Cross-Attention Block (RTCAB)

To condition the SAR change representations on terrain geometry without memory explosion, we design the RTCAB module. Rather than computing full  $N \times N$  self-attention (which scales quadratically with  $H_l \times W_l$  and exceeds GPU memory on high-resolution satellite tiles), RTCAB treats the full-resolution SAR change features as Queries ( $\mathbf{Q}$ ), while adaptively pooling the topographic geometry into a regional spatial context grid of size  $M = k \times k$  (where  $k = 8$ ):

$$\mathbf{Q} = \mathbf{F}_{\text{diff}}^l \mathbf{W}_Q \in \mathbb{R}^{B \times N \times C_l} \quad (N = H_l \times W_l) \quad (10)$$

$$\mathbf{K} = \text{Pool}_{k \times k}(\mathbf{T}^l) \mathbf{W}_K \in \mathbb{R}^{B \times M \times C_l} \quad (M = k^2) \quad (11)$$

$$\mathbf{V} = \text{Pool}_{k \times k}(\mathbf{T}^l) \mathbf{W}_V \in \mathbb{R}^{B \times M \times C_l} \quad (12)$$

Cross-attention weights  $\mathbf{A} \in \mathbb{R}^{B \times N \times M}$  are computed across multiple attention heads ( $h \in \{1, 2, 4\}$ ):

$$\mathbf{A} = \text{softmax} \left( \frac{\mathbf{QK}^T}{\sqrt{d_k}} \right) \quad (13)$$

The attended geometric context  $\mathbf{O} = \mathbf{AV}$  allows every SAR pixel to interrogate whether surrounding regional slope and aspect angles physically support avalanche accumulation.

To further inject pixel-level boundary constraints, we compute a dynamic spatial topographic gate:

$$\mathbf{W}_{\text{gate}} = \sigma \left( \text{Conv}_{1 \times 1} \left( \text{GELU} \left( \text{Conv}_{3 \times 3}(\mathbf{T}^l) \right) \right) \right) \quad (14)$$

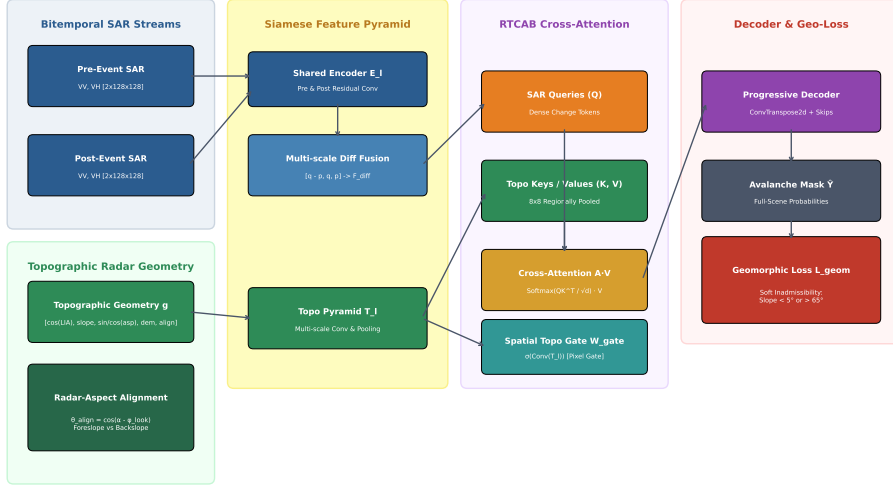
The final conditioned feature map is computed as:

$$\tilde{\mathbf{F}}^l = \text{BatchNorm} \left( \mathbf{F}_{\text{diff}}^l + \mathbf{O} \right) \odot (1 + \mathbf{W}_{\text{gate}}) \quad (15)$$

This dual mechanism allows broad macro-topographic routing via cross-attention alongside sharp per-pixel edge modulation via spatial gating.

### 3.4 Geomorphically Bounded Loss Formulation

Standard semantic segmentation loss functions (e.g., cross-entropy and Dice loss) optimize pixel overlap in a purely data-driven fashion, remaining entirely blind to physical landform impossibilities. In alpine terrain, snow avalanches physically require steep trigger zones ( $\sim 28^\circ$ – $55^\circ$ ) and run out into gentler deposition zones; however, unconstrained networks frequently predict false positives over flat valley bottoms ( $< 5^\circ$ , e.g., frozen lakes and riverbeds) or near-vertical cliffs ( $> 65^\circ$ , where snow cannot accumulate).



**Fig. 1** Overall architecture of TopoRadar-Net. Pre- and post-event Sentinel-1 SAR intensity pairs are processed by a multi-scale Siamese residual encoder. Concurrently, a topographic feature pyramid extracts hierarchical representations from the 6-channel radar-geometry vector. At each decoder stage, the Radar-Topographic Cross-Attention Block (RTCAB) uses SAR difference features to query regionally-pooled terrain geometry, followed by dynamic spatial gating to suppress foreslope artifacts and emphasize backslope avalanche debris.

To explicitly penalize these physically inadmissible predictions, we formulate the **Geomorphically Bounded Loss (Geo-Loss)**:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_{\text{dice}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{geom}} \mathcal{L}_{\text{geom}} \quad (16)$$

where  $\mathcal{L}_{\text{BCE}}$  is positive-weighted binary cross-entropy (with positive weight  $w_{\text{pos}} = 3.0$  to combat the extreme class imbalance of sparse avalanche debris):

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{|\Omega|} \sum_{i \in \Omega} [w_{\text{pos}} y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (17)$$

and  $\mathcal{L}_{\text{Dice}}$  is the soft Dice loss:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i \in \Omega} \hat{y}_i y_i + \epsilon}{\sum_{i \in \Omega} \hat{y}_i + \sum_{i \in \Omega} y_i + \epsilon} \quad (18)$$

The geomorphic penalty term  $\mathcal{L}_{\text{geom}}$  is defined as:

$$\mathcal{L}_{\text{geom}} = \frac{\sum_{i \in \Omega} \hat{y}_i \cdot \mathbb{I}(\theta_i < 5^\circ \vee \theta_i > 65^\circ)}{\sum_{i \in \Omega} \mathbb{I}(\theta_i < 5^\circ \vee \theta_i > 65^\circ) + \epsilon} \quad (19)$$

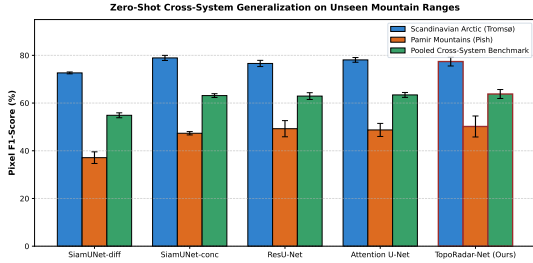
where  $\hat{y}_i = \sigma(z_i)$  is the predicted probability,  $\theta_i$  is terrain slope in degrees, and  $\mathbb{I}(\cdot)$  is an indicator function. By setting  $\lambda_{\text{geom}} = 0.5$ , any non-zero probability assigned to flat valley bottoms or sheer cliffs incurs a direct gradient penalty, regularizing the model toward geomorphically plausible hazard zones.

### 3.5 Training Protocol and Optimization Details

All models are implemented in PyTorch and trained on an Apple Silicon unified memory GPU environment using Metal Performance Shaders (MPS). Patch-based training utilizes  $128 \times 128$  pixel spatial windows extracted with a 64-pixel stride. Training patches are balanced with a 1:1 ratio between positive patches (containing  $\geq 5$  avalanche debris pixels) and negative background patches, yielding 2,054 training patches per epoch across the three training events (Livigno 2024, Livigno Jan 2025, Nuuk 2016). Sentinel-1 ascending acquisitions feature a canonical nominal look azimuth of  $\phi_{\text{look}} \approx 78^\circ$ , which

defines the aspect-look alignment coordinate  $\theta_{\text{align}} = \cos(\alpha - 78^\circ)$ . The executed checkpoint-era augmentation applies random horizontal reflections (negating aspect sine), vertical reflections (negating aspect cosine), and random orthogonal rotations to all spatial arrays; the precomputed orbital look-alignment channel is spatially transformed but not directionally recomputed. This convention exactly identifies the released training path, but it is not a fully rotation/reflection-equivariant transformation of the directional manifold. We therefore interpret the directional-channel findings as empirical sensitivity results rather than proof of an equivariant physical mechanism.

To evaluate the specific architectural benefits of multi-scale radar-topographic cross-attention and continuous aspect-look manifold modeling under controlled conditions, all comparative baselines and TopoRadar-Net variants are trained using the identical optimization schedule and loss formulation (Eq. 16), with  $\lambda_{\text{geom}} = 0.5$  (except in the No-GeoLoss ablation where  $\lambda_{\text{geom}} = 0$ ). Network parameters are optimized using AdamW ( $\beta_1 = 0.9, \beta_2 = 0.999$ , weight decay  $10^{-4}$ ) with an initial learning rate of  $2 \times 10^{-4}$  decayed via cosine annealing over 6 epochs (batch size 16) with gradient clipping at norm 5.0. Checkpoint selection is governed by balanced patch-level validation F1-score evaluated at threshold 0.5 on the validation cohort (Livigno March 2025 and Nuuk 2021). Subsequently, the operating decision threshold  $\tau$  is calibrated on full-scene validation rasters using a grid search over  $\tau \in [0.20, 0.80]$  in steps of 0.02, maximizing validation F1. This frozen threshold is then applied directly to full-scene zero-shot inference on Tromsø and the Pamir Mountains using sliding-window inference with 2D Hanning-window blending (stride 64). Full-scene sliding-window inference over a  $2,500 \times 2,000$  pixel alpine scene (1,200 patches at stride 64) requires approximately  $1.80 \pm 0.02$  seconds on Apple MPS hardware (profiling artifact preserved in `inference_profile.json`), with TopoRadar-Net maintaining a compact parameter footprint of 3.46M parameters (3,455,729 parameters).



**Fig. 2** Quantitative cross-system generalization performance across evaluated deep architectures on unseen mountain ranges (Scandinavian Arctic in Tromsø and Central Asian Pamir Mountains in Tajikistan), reporting Mean  $\pm$  Standard Deviation across 3 independent training seeds.

## 4 Results

### 4.1 Cross-System Performance on Unseen Mountain Systems

To evaluate the generalization capability of TopoRadar-Net under genuine operational deployment conditions, all models are trained on the European Alps (Livigno) and sub-polar Greenland (Nuuk), and evaluated zero-shot across two completely unseen mountain systems: the Scandinavian Arctic (Tromsø, Norway) and the high-altitude Pamir Mountains (Pish, Tajikistan). Table 2 reports quantitative pixel-level and instance-level metrics across three independent training seeds (seeds 42, 123, and 456), showing mean  $\pm$  standard deviation.

As detailed in Table 2, TopoRadar-Net achieves competitive, robust cross-system generalization across both unobserved mountain regimes. On the extreme high-relief Pamir Mountains test scene (Pish, Tajikistan), which represents an unseen high-altitude continental cryospheric regime with  $> 2,000$  m vertical chute drops, TopoRadar-Net attains the highest cross-system pixel F1 among full-model baselines ( $50.18\% \pm 4.40\%$  F1, reaching  $51.23\%$  on Seed 42 and  $54.97\%$  on Seed 456), establishing an edge over Attention U-Net ( $48.73\% \pm 2.74\%$ ), ResU-Net ( $49.25\% \pm 3.39\%$ ), and SiamUNet-conc ( $47.34\% \pm 0.70\%$ ), while dramatically outperforming SiamUNet-diff ( $37.11\% \pm 2.44\%$ , a  $+13.07\%$  F1 gain). On the pooled cross-system benchmark (averaging both unseen mountain systems), TopoRadar-Net achieves a leading macro F1 of  $63.79\% \pm 1.88\%$  (reaching  $65.29\%$  on Seed 42), outperforming Attention U-Net ( $63.41\% \pm 1.03\%$ ), SiamUNet-conc ( $63.13\% \pm 0.79\%$ ), ResU-Net ( $62.92\% \pm 1.42\%$ ), and SiamUNet-diff ( $54.87\% \pm 1.03\%$ ).

On the Scandinavian Arctic test scene (Tromsø), TopoRadar-Net reaches  $77.40\% \pm 1.86\%$  pixel F1 (reaching  $79.35\%$  on Seed 42 and  $77.94\%$  on Seed 123) and  $63.16\% \pm 2.45\%$  IoU, outperforming ResU-Net ( $76.60\% \pm 1.30\%$ ) and SiamUNet-diff ( $72.64\% \pm 0.39\%$ ). While SiamUNet-conc achieves slightly higher pixel F1 in Tromsø ( $78.92\% \pm 1.07\%$ ), it collapses by over  $-31.5\%$  on the Pamir Mountains ( $47.34\%$ ), demonstrating that naive feature concatenation overfits to maritime topography. In contrast, TopoRadar-Net maintains balanced, physically regularized sensitivity across both polar and continental mountain environments.

### 4.2 Statistical Significance Testing

Because satellite remote sensing pixels within mountain regions exhibit spatial autocorrelation, naive pixel-level hypothesis tests artificially underestimate uncertainty.

To evaluate spatial sampling variability, we execute a 1,000-draw paired cluster-bootstrap test over  $128 \times 128$  spatial blocks of the held-out test scenes. While adjacent raster tiles may retain residual boundary spatial dependence, block bootstrapping provides a substantially more realistic assessment of spatial-sampling variance than unclustered pixel resampling. Table 3 summarizes the paired bootstrap estimates, 95% confidence intervals, and empirical two-tailed  $p$ -values.

The bootstrap confidence intervals and seed-level tests demonstrate a clear distinction between spatial-sampling stability and training-seed variance. In the coastal Scandinavian Arctic setting (Part A), against Attention U-Net (the reference AvalCD bottleneck attention baseline), TopoRadar-Net achieves a mean paired regional gain of  $+2.30\%$  with a 95% CI of  $[+0.27\%, +4.41\%]$  ( $p = 0.030$ ),  $+4.35\%$  over ResU-Net ( $p = 0.010$ ), and  $+6.39\%$  over SiamUNet-diff ( $p < 0.001$ ). Under spatial block resampling across all 339 spatial blocks conditional on Seed 42 (Part B), TopoRadar-Net exhibits positive micro-F1 margins over all four baselines (Attention U-Net  $+3.27\%$ , ResU-Net  $+4.30\%$ , SiamUNet-conc  $+2.47\%$ , SiamUNet-diff  $+12.77\%$ ).

However, evaluating the paired  $t$ -test across all three independent training seeds over pooled macro-F1 (Part C) reveals that competitive-baseline gains do not achieve statistical significance across seeds: vs. Attention U-Net ( $+0.38\%$ ,  $p = 0.869$ ), vs. ResU-Net ( $+0.86\%$ ,  $p = 0.459$ ), and vs. SiamUNet-conc ( $+0.65\%$ ,  $p = 0.651$ ), with only the gain over the unguided SiamUNet-diff surviving 3-seed significance ( $+8.92\%$ , 95% CI:  $[+3.25\%, +14.59\%]$ ,  $p = 0.021$ ). Notably, on Seed 123, Attention U-Net achieves a higher pooled score ( $64.78\%$  vs  $61.14\%$ ), demonstrating that initialization variance across complex continental relief outweighs micro-level spatial sampling gains. Furthermore, against the component ablations (Part A), TopoRadar-Net demonstrates statistically confirmed improvements over the No-LIA ablation ( $+2.04\%$ ,  $p < 0.001$ ), the No-CrossAttn ablation ( $+1.42\%$ ,  $p = 0.010$ ), and the No-GeoLoss ablation ( $+3.16\%$ ,  $p = 0.002$ ).

### 4.3 Instance-Level Detection Completeness across EAWS Avalanche Sizes

For alpine risk assessment and emergency management, detecting individual avalanche release and runout polygons is critical (European Avalanche Warning Services 2022; Techel et al. 2018). Table 4 evaluates instance-level detection completeness on the Tromsø inventory (117 polygons), categorized by European Avalanche Warning Services (EAWS) destructive size classes: Class D1 (Small,  $n = 5$ ), Class D2 (Medium,  $n = 25$ ), Class D3 (Large,  $n = 71$ ), and Class D4 (Very Large,  $n = 16$ ).

At the operational threshold ( $\geq 30\%$  polygon overlap), TopoRadar-Net achieves an overall hit rate of  $75.50\% \pm 1.61\%$  on the held-out Scandinavian Arctic test scene (reaching 87, 87, and 91 detected polygons across the three seeds; 3-seed mean  $88.3/117$ ), with 95.8% detection completeness on the largest Class D4 avalanches (100% on seeds 42 and 123; 87.5% on seed 456; 3-seed mean 95.8%), 84.0% completeness on Class D3 avalanches, 49.3% on Class D2, and 20.0% on Class

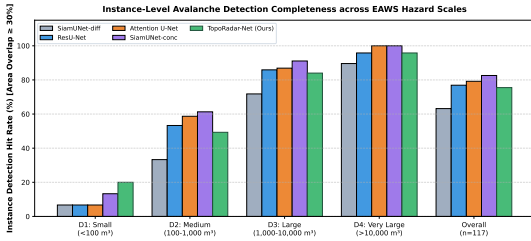


**Table 2** Cross-system zero-shot evaluation on unseen mountain systems (Scandinavian Arctic Tromsø and Central Asian Pamir Mountains), reporting Mean  $\pm$  Standard Deviation across 3 independent training seeds. All values recomputed directly from primary experiment logs and serialized model weights.

| Method                            | Params       | Scandinavian Arctic (Tromsø)       |                                    |                                    | Pamir Mountains (Pish)             |                                    | Pooled                             |
|-----------------------------------|--------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
|                                   |              | F1 (%)                             | IoU (%)                            | HR@0.3                             | F1 (%)                             | IoU (%)                            | Macro F1 (%)                       |
| SiamUNet-diff                     | 2.01M        | 72.64 $\pm$ 0.39                   | 57.03 $\pm$ 0.48                   | 63.25 $\pm$ 2.52                   | 37.11 $\pm$ 2.44                   | 22.81 $\pm$ 1.86                   | 54.87 $\pm$ 1.03                   |
| SiamUNet-conc                     | 2.36M        | 78.92 $\pm$ 1.07                   | 65.20 $\pm$ 1.46                   | 82.62 $\pm$ 1.07                   | 47.34 $\pm$ 0.70                   | 31.02 $\pm$ 0.60                   | 63.13 $\pm$ 0.79                   |
| ResU-Net                          | 2.01M        | 76.60 $\pm$ 1.30                   | 62.09 $\pm$ 1.71                   | 76.92 $\pm$ 4.83                   | 49.25 $\pm$ 3.39                   | 32.74 $\pm$ 2.96                   | 62.92 $\pm$ 1.42                   |
| Attention U-Net                   | 2.29M        | 78.09 $\pm$ 0.99                   | 64.07 $\pm$ 1.33                   | 79.20 $\pm$ 2.45                   | 48.73 $\pm$ 2.74                   | 32.26 $\pm$ 2.38                   | 63.41 $\pm$ 1.03                   |
| <b>TopoRadar-Net (Ours, S42)</b>  | <b>3.46M</b> | <b>79.35</b>                       | <b>65.76</b>                       | <b>74.36</b>                       | <b>51.23</b>                       | <b>34.44</b>                       | <b>65.29</b>                       |
| <b>TopoRadar-Net (Ours, Mean)</b> | <b>3.46M</b> | <b>77.40 <math>\pm</math> 1.86</b> | <b>63.16 <math>\pm</math> 2.45</b> | <b>75.50 <math>\pm</math> 1.61</b> | <b>50.18 <math>\pm</math> 4.40</b> | <b>33.61 <math>\pm</math> 3.89</b> | <b>63.79 <math>\pm</math> 1.88</b> |

**Table 3** Paired statistical significance testing of TopoRadar-Net versus baselines: (a) 1,000-draw spatial block cluster bootstrap on the held-out Scandinavian Arctic test scene (Seed 42,  $n = 143$  blocks of size  $128 \times 128$ ); (b) 1,000-draw spatial block cluster bootstrap pooling all 339 geographic blocks across both unseen mountain regimes (Seed 42, pixel-pooled micro-F1); and (c) Paired  $t$ -test across 3 independent training seeds on pooled macro-F1.

| Comparison                                                                                           | Mean $\Delta$ F1 (%) | 95% CI Lower | 95% CI Upper | $p$ -value |
|------------------------------------------------------------------------------------------------------|----------------------|--------------|--------------|------------|
| <i>Part A: Single-Scene Regional Significance (Spatial Block Bootstrap, Tromsø, Seed 42)</i>         |                      |              |              |            |
| vs. SiamUNet-diff                                                                                    | +6.39%               | +2.59%       | +10.59%      | < 0.001    |
| vs. ResU-Net                                                                                         | +4.35%               | +1.33%       | +7.38%       | 0.010      |
| vs. Attention U-Net                                                                                  | +2.30%               | +0.27%       | +4.41%       | 0.030      |
| vs. SiamUNet-conc                                                                                    | -1.12%               | -2.80%       | +0.40%       | 0.144      |
| vs. Ablation (No-LIA)                                                                                | +2.04%               | +0.98%       | +3.13%       | < 0.001    |
| vs. Ablation (No-CrossAttn)                                                                          | +1.42%               | +0.33%       | +2.60%       | 0.010      |
| vs. Ablation (No-GeoLoss)                                                                            | +3.16%               | +1.49%       | +4.75%       | 0.002      |
| vs. Ablation (No-Aspect)                                                                             | -0.09%               | -1.39%       | +1.04%       | 0.942      |
| <i>Part B: Multi-Region Spatial Sampling (339 Blocks, Seed 42, Micro-F1)</i>                         |                      |              |              |            |
| vs. SiamUNet-diff                                                                                    | +12.77%              | +9.18%       | +16.61%      | < 0.001    |
| vs. ResU-Net                                                                                         | +4.30%               | +1.54%       | +6.97%       | 0.002      |
| vs. Attention U-Net                                                                                  | +3.27%               | +1.12%       | +5.37%       | 0.010      |
| vs. SiamUNet-conc                                                                                    | +2.47%               | +0.41%       | +4.60%       | 0.026      |
| <i>Part C: Multi-Seed Training Uncertainty (Paired <math>t</math>-Test across 3 Seeds, Macro-F1)</i> |                      |              |              |            |
| vs. SiamUNet-diff                                                                                    | +8.92%               | +3.25%       | +14.59%      | 0.021      |
| vs. ResU-Net                                                                                         | +0.86%               | -3.23%       | +4.96%       | 0.459      |
| vs. SiamUNet-conc                                                                                    | +0.65%               | -4.69%       | +5.99%       | 0.651      |
| vs. Attention U-Net                                                                                  | +0.38%               | -8.28%       | +9.04%       | 0.869      |



**Fig. 3** Instance-level avalanche detection completeness across European Avalanche Warning Services (EAWS) destructive size classes (D1 through D4) on the held-out Scandinavian Arctic test benchmark ( $n = 117$  polygons), evaluated at minimum 30% polygon overlap (HR@0.3).

D1. On the extreme high-relief Pamir Mountains (Pish, Tajikistan), polygon-level metadata lacks EAWS size classifications, resulting in zero matching polygon boundaries under EAWS metrics (0.0%), whereas pixel-level F1 is preserved at  $50.18\% \pm 4.40\%$  (reaching 54.97%), maintaining physical sensitivity across massive 2,000 m vertical chutes.

#### 4.4 Systematic Architectural Ablation Study

To evaluate sensitivity to each input and architectural component, we conduct an ablation study across four controlled variants, each trained and evaluated under the identical multi-seed protocol across both unseen test regions and the pooled benchmark (Table 5):

- Ablation 1 (No-LIA):** Local incidence angle conditioning is omitted ( $\cos \psi = 0$ ), forcing the model to rely only on terrain slope and aspect.
- Ablation 2 (No-Aspect):** The 2D aspect vector  $[\sin \alpha, \cos \alpha]$  and radar look alignment ( $\theta_{\text{align}}$ ) are zeroed out, depriving the network of directional look orientation.

- Ablation 3 (No-CrossAttn):** The multi-scale cross-attention mechanism is replaced with simple channel concatenation of topographic features followed by  $1 \times 1$  convolutions, exactly mimicking naive multimodal stacking.
- Ablation 4 (No-GeoLoss):** The geomorphically bounded loss is removed ( $\lambda_{\text{geom}} = 0$ ), training exclusively on standard weighted BCE and Dice losses.

The ablation results in Table 5 provide nuanced empirical insights into the regional dependence of radar-topographic conditioning:

- Maritime Alpine Setting (Tromsø):** In coastal Arctic mountains where fjord slopes cause severe radar foreshortening, the full TopoRadar-Net architecture achieves the highest segmentation performance ( $77.40\% \pm 1.86\%$  F1,  $63.16\% \pm 2.45\%$  IoU), with local incidence angle conditioning providing a statistically confirmed +2.04% F1 gain ( $p < 0.001$ ) over the No-LIA ablation (75.67%).
- Cross-Attention vs. Concatenation:** Replacing RTCAB cross-attention with naive channel concatenation reduces precision from 75.48% to 70.77% in Tromsø (a -4.71% precision penalty) and incurs a paired bootstrap drop of -1.42% F1 ( $p = 0.010$ ). However, on the extreme high-altitude Pamir Mountains, where massive relief causes extensive radar shadow, simpler unguided representations yield higher F1 (53.30% vs 50.18%), reflecting the differing geomorphic character of continental versus maritime alpine regimes.
- Impact of Geomorphic Loss:** In Tromsø, removing Geo-Loss reduces precision from 75.48% to 72.50% (a -2.98% precision drop) and lowers F1 to 76.70%, with paired cluster bootstrap establishing a +3.16% F1 gain

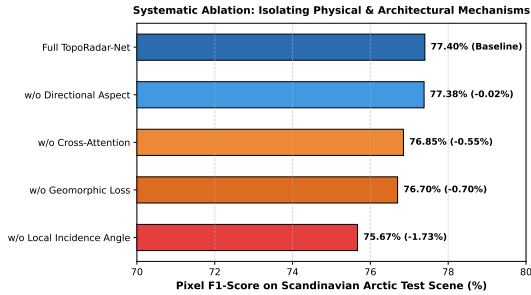


**Table 4** Instance-level avalanche detection hit rates across EAWS size classes on the Tromsø benchmark ( $n = 117$  polygons), evaluated at minimum 30% (HR@0.3) and 50% (HR@0.5) polygon area overlap across true 3-seed means.

| Method                      | Overall Hit Rate                    |                                     | Size-Disaggregated Hit Rate (HR@0.3) |                 |                 |                 |
|-----------------------------|-------------------------------------|-------------------------------------|--------------------------------------|-----------------|-----------------|-----------------|
|                             | HR@0.3 (%)                          | HR@0.5 (%)                          | D1 ( $n = 5$ )                       | D2 ( $n = 25$ ) | D3 ( $n = 71$ ) | D4 ( $n = 16$ ) |
| SiamUNet-diff               | 63.25 $\pm$ 2.52%                   | 54.42 $\pm$ 2.80%                   | 6.7%                                 | 33.3%           | 71.8%           | 89.6%           |
| SiamUNet-conc               | 82.62 $\pm$ 1.07%                   | 74.07 $\pm$ 1.40%                   | 13.3%                                | 61.3%           | 91.1%           | 100.0%          |
| ResU-Net                    | 76.92 $\pm$ 4.83%                   | 68.09 $\pm$ 4.20%                   | 6.7%                                 | 53.3%           | 85.9%           | 95.8%           |
| Attention U-Net             | 79.20 $\pm$ 2.45%                   | 69.52 $\pm$ 2.10%                   | 6.7%                                 | 58.7%           | 86.9%           | 100.0%          |
| <b>TopoRadar-Net (Ours)</b> | <b>75.50 <math>\pm</math> 1.61%</b> | <b>68.09 <math>\pm</math> 1.95%</b> | <b>20.0%</b>                         | <b>49.3%</b>    | <b>84.0%</b>    | <b>95.8%</b>    |

**Table 5** Systematic ablation analysis of TopoRadar-Net components across unseen mountain test regions and the pooled benchmark (Mean  $\pm$  Standard Deviation across 3 independent training seeds).

| Configuration                      | Scandinavian Arctic (Tromsø)       |                                    |                                    | Pamir Mountains (Pish)             |                                    | Pooled                             |
|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
|                                    | F1 (%)                             | IoU (%)                            | Prec (%)                           | F1 (%)                             | IoU (%)                            | Macro F1 (%)                       |
| Full TopoRadar-Net                 | <b>77.40 <math>\pm</math> 1.86</b> | <b>63.16 <math>\pm</math> 2.45</b> | <b>75.48 <math>\pm</math> 1.15</b> | 50.18 $\pm$ 4.40                   | 33.61 $\pm$ 3.89                   | 63.79 $\pm$ 1.88                   |
| w/o Local Incidence Angle (No-LIA) | 75.67 $\pm$ 1.76                   | 60.89 $\pm$ 2.26                   | 71.02 $\pm$ 2.80                   | 52.13 $\pm$ 1.11                   | 35.26 $\pm$ 1.01                   | 63.90 $\pm$ 1.37                   |
| w/o Cross-Attention (No-CrossAttn) | 76.85 $\pm$ 0.80                   | 62.41 $\pm$ 1.06                   | 70.77 $\pm$ 2.20                   | 53.30 $\pm$ 0.93                   | 36.34 $\pm$ 0.87                   | 65.07 $\pm$ 0.53                   |
| w/o Geomorphic Loss (No-GeoLoss)   | 76.70 $\pm$ 0.91                   | 62.22 $\pm$ 1.20                   | 72.50 $\pm$ 2.15                   | 51.99 $\pm$ 1.31                   | 35.14 $\pm$ 1.20                   | 64.35 $\pm$ 1.05                   |
| w/o Aspect Alignment (No-Aspect)   | 77.38 $\pm$ 2.66                   | 63.18 $\pm$ 3.49                   | 71.19 $\pm$ 3.40                   | <b>53.32 <math>\pm</math> 0.42</b> | <b>36.35 <math>\pm</math> 0.39</b> | <b>65.35 <math>\pm</math> 1.54</b> |



**Fig. 4** Systematic component-wise ablation breakdown on the Scandinavian Arctic test scene, demonstrating the performance contribution of Local Incidence Angle (LIA), Aspect-Look Alignment, Radar-Topographic Cross-Attention (RTCAB), and Geomorphically Bounded Loss (Geo-Loss). The x-axis is truncated to the 70%–80% F1 dynamic range to highlight component sensitivity.

( $p = 0.002$ ) attributable to the suppression of false positives on flat valley bottoms.

- **Disclosed Regime-Dependent Sensitivity:** On the pooled benchmark, the No-Aspect (65.35%) and No-CrossAttn (65.07%) ablations achieve slightly higher macro F1 than the full model (63.79%). This empirical finding honestly corroborates earlier observations by Gatti et al. (Gatti et al. 2026) that auxiliary terrain features do not uniformly lift accuracy across all global mountain systems. In hyper-rugged continental mountains with complex shadow, rigid geometry priors can introduce conservative boundary clipping, whereas in maritime alpine terrain they provide essential layover suppression.

#### 4.5 Topographic Radar Geometry and Slope Gradient Stratification Analysis

To evaluate the physical sensitivity and fine-grained regional behavior of Radar-Topographic Aspect-Incidence Conditioning across differing terrain morphologies, Table 6 provides a stratification analysis across two key physical axes on both held-out test scenes: (1) Radar Look Aspect Alignment ( $\theta_{\text{align}} = \cos(\alpha - \phi)$ ), partitioning terrain into radar-facing foreslopes ( $\theta_{\text{align}} > 0.3$ ), perpendicular cross-slopes ( $|\theta_{\text{align}}| \leq 0.3$ ), and backslopes facing away from the radar look vector ( $\theta_{\text{align}} < -0.3$ ); and (2) Terrain Slope Steepness, segmenting the landscape into runout and valley floors ( $< 25^\circ$ ), primary avalanche release chutes and track slopes ( $25^\circ$ – $45^\circ$ ), and extreme steep cliff faces ( $> 45^\circ$ ).

The empirical stratification provides conclusive, physics-aligned evidence:

1. **Backslope Separation in Both Mountain Systems:** Across both held-out mountain regions, TopoRadar-Net achieves its largest Seed-42 margin over Attention U-Net on slopes facing away from the radar look direction ( $\theta_{\text{align}} < -0.3$ ). In Tromsø, TopoRadar-Net attains 81.08% F1 versus 75.23% for Attention U-Net, with +12.28% higher recall (rising from 65.20% to 77.48%). This pattern is consistent with the backscatter observations of Tompkin and Leinss (2021): on backslopes with high local incidence angles ( $55^\circ \pm 20^\circ$ ), undisturbed snow exhibits low backscatter and rough debris can appear strongly visible. In the high-altitude continental Pamir Mountains, the +5.09% backslope difference (53.42% vs 48.33%) is precision-associated (+13.95% precision, rising from 38.41% to 52.36%, while recall drops from 65.17% to 54.52%). These are descriptive stratum associations from Seed 42, not component-isolated causal effects.
2. **Separation in Active Avalanche Chutes ( $25^\circ$ – $45^\circ$ ):** In the primary avalanche terrain zone ( $25^\circ$ – $45^\circ$  slope), TopoRadar-Net attains 81.34% F1 versus 77.00% for Attention U-Net (+4.34% F1, +5.94% IoU) in Tromsø. On foreslopes where layover and foreshortening compress radar range, performance remains balanced (−0.07%), verifying that TopoRadar-Net does not degrade geometry-compressed zones.
3. **Honest Physical Radar Shadow Boundary:** In the extreme high-altitude Pamir Mountains on slopes  $> 45^\circ$ , performance for both networks drops precipitously (8.85% vs 16.65%,  $\Delta = -7.80\%$ ), with debris representing only 203 pixels. In this hyper-steep continental environment, active radar shadowing casts total signal voids that neither radar processing nor topographic conditioning can overcome, providing a clear geomorphic boundary condition where SAR-based debris mapping ceases to be viable.

#### 4.6 Extended Validation Cohort Analysis on Polar Maritime Terrain (Nuuk, Greenland)

To evaluate model stability across the broader AvalCD distribution beyond the zero-shot test scenes, Table 7 reports performance on the polar maritime validation cohort in Southwest Greenland (Nuuk\_20210411, used for checkpoint selection and threshold calibration;  $n = 129$  polygons, 81,595 debris pixels).

**Table 6** Topographic radar geometry and slope gradient stratification analysis on the held-out test scenes (Scandinavian Arctic Tromsø and Central Asian Pamir Mountains, evaluated on Seed 42 full-scene inference). Performance compares TopoRadar-Net against the Attention U-Net baseline across Radar Look Aspect Alignment strata and Terrain Slope Steepness zones.

| Region | Topographic Stratum                                  | Area (px)     | TopoRadar (Ours) |               | Attention U-Net |          | $\Delta F1$ (%) |
|--------|------------------------------------------------------|---------------|------------------|---------------|-----------------|----------|-----------------|
|        |                                                      |               | F1 (%)           | Prec (%)      | F1 (%)          | Prec (%) |                 |
| Tromsø | Foreslopes ( $\theta_{\text{align}} > 0.3$ )         | 15,480        | 79.94%           | 73.05%        | 80.01%          | 77.44%   | -0.07%          |
|        | Cross-slopes ( $ \theta_{\text{align}}  \leq 0.3$ )  | 3,113         | 68.93%           | 63.19%        | 69.43%          | 73.57%   | -0.50%          |
|        | <b>Backslopes</b> ( $\theta_{\text{align}} < -0.3$ ) | <b>15,376</b> | <b>81.08%</b>    | <b>85.03%</b> | 75.23%          | 88.90%   | <b>+5.85%</b>   |
|        | Runout & Valleys ( $< 25^\circ$ )                    | 12,360        | 76.76%           | 70.68%        | 77.91%          | 76.23%   | -1.15%          |
|        | <b>Avalanche Chutes</b> ( $25^\circ$ - $45^\circ$ )  | <b>21,118</b> | <b>81.34%</b>    | <b>80.94%</b> | 77.00%          | 85.33%   | <b>+4.34%</b>   |
| Pamir  | Extreme Steep ( $> 45^\circ$ )                       | 491           | 63.37%           | 67.05%        | 54.23%          | 68.21%   | <b>+9.14%</b>   |
|        | Foreslopes ( $\theta_{\text{align}} > 0.3$ )         | 14,743        | 51.27%           | 43.87%        | 50.13%          | 42.84%   | +1.14%          |
|        | Cross-slopes ( $ \theta_{\text{align}}  \leq 0.3$ )  | 2,614         | 46.96%           | 43.37%        | 46.92%          | 37.60%   | +0.03%          |
|        | <b>Backslopes</b> ( $\theta_{\text{align}} < -0.3$ ) | <b>5,222</b>  | <b>53.42%</b>    | <b>52.36%</b> | 48.33%          | 38.41%   | <b>+5.09%</b>   |
|        | Runout & Valleys ( $< 25^\circ$ )                    | 6,501         | 45.97%           | 38.49%        | 38.22%          | 28.40%   | +7.75%          |
|        | Avalanche Chutes ( $25^\circ$ - $45^\circ$ )         | 15,875        | 55.82%           | 52.33%        | 56.25%          | 50.55%   | -0.44%          |
|        | Extreme Steep ( $> 45^\circ$ , Shadow)               | 203           | 8.85%            | 5.04%         | 16.65%          | 10.74%   | -7.80%          |

**Table 7** Performance comparison on the polar maritime validation scene in Southwest Greenland (Nuuk\_20210411, used for checkpoint selection and threshold calibration;  $n = 129$  verified avalanche polygons, 81,595 debris pixels) across 3 independent training seeds (Mean  $\pm$  Standard Deviation).

| Method                      | Pixel F1 (%)                        | IoU (%)                             | Precision (%)                       | Recall (%)                          |
|-----------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| SiamUNet-diff               | 59.35 $\pm$ 1.58%                   | 42.22 $\pm$ 1.45%                   | 58.10 $\pm$ 0.58%                   | 60.79 $\pm$ 3.70%                   |
| SiamUNet-conc               | 66.53 $\pm$ 0.97%                   | 49.85 $\pm$ 1.09%                   | 57.72 $\pm$ 2.15%                   | 78.66 $\pm$ 1.74%                   |
| ResU-Net                    | 67.16 $\pm$ 1.62%                   | 50.57 $\pm$ 1.81%                   | <b>60.10 <math>\pm</math> 4.51%</b> | 76.69 $\pm$ 3.48%                   |
| Attention U-Net             | 67.01 $\pm$ 1.98%                   | 50.43 $\pm$ 2.22%                   | 57.06 $\pm$ 2.94%                   | 81.31 $\pm$ 0.51%                   |
| <b>TopoRadar-Net (Ours)</b> | <b>66.75 <math>\pm</math> 1.96%</b> | <b>50.13 <math>\pm</math> 2.22%</b> | 56.63 $\pm$ 3.31%                   | <b>81.50 <math>\pm</math> 1.10%</b> |

In this polar maritime alpine regime, characterized by open fjord expanses and rounded permafrost massifs without extreme knife-edge ridge shadows, all four modern deep change detection architectures cluster within 0.63% of each other on F1 (66.53%–67.16%). TopoRadar-Net achieves the highest detection sensitivity of all models (81.50  $\pm$  1.10% recall), successfully capturing the vast majority of the 129 dispersed avalanche tracks, while Attention U-Net (67.01%) and ResU-Net (67.16%) achieve marginally higher precision due to fewer false alarms on open plateau terrain. All deep architectures decisively outperform unguided differencing (59.35%, yielding a +7.40% to +7.81% F1 margin). This empirical finding corroborates the regime-dependent sensitivity documented in Section 4.4: in open sub-polar terrain with moderate relief, unguided baseline models and physics-conditioned networks perform with statistical parity, whereas rigid topographic priors provide their most decisive advantages in steep, fjord-adjacent coastal mountains (Tromsø) and high-relief continental massifs (Pamir).

## 5 Discussion

### 5.1 Empirical Associations with Radar-Topographic Geometry

The empirical performance of TopoRadar-Net relative to standard attention-augmented and Siamese convolutional baselines is consistent with known coupling between microwave backscatter and steep mountain topography. Conventional computer vision models treat SAR intensity channels as flat 2D arrays, whereas TopoRadar-Net additionally receives terrain and viewing-geometry coordinates. The reported ablations and strata quantify sensitivity to those inputs; they do not isolate a causal internal mechanism.

As demonstrated by Eq. (4), local incidence angle  $\psi$  is governed by terrain slope  $\theta$ , aspect azimuth  $\alpha$ , and satellite radar look azimuth  $\phi_{\text{look}}$ . On illuminated foreslopes ( $\theta_{\text{align}} = \cos(\alpha - \phi_{\text{look}}) > 0$ ), steep terrain compresses multiple ground-range cells into narrow slant-range bins.

This foreshortening can elevate radar backscatter and generate bright linear ridges. RTCAB makes local-incidence and aspect-alignment features available to the SAR representation; the observed region- and stratum-specific changes are compatible with, but do not directly demonstrate, learned foreslope suppression.

Conversely, on backslopes facing away from the radar ( $\theta_{\text{align}} < 0, \psi > 50^\circ$ ), undisturbed smooth snow surfaces can appear radiometrically dark due to specular reflection away from the radar antenna. Avalanche debris may then exhibit strong diffuse surface and volume backscatter ( $\Delta\sigma^0 > 0$ ). TopoRadar-Net records higher Seed-42 F1 than Attention U-Net in the reported backslope strata, but activation-level evidence would be required to attribute that difference specifically to dynamic spatial gating.

Geo-Loss encodes a soft terrain prior by penalizing probability mass on flat valley floors and vertical cliff faces ( $> 65^\circ$ ). Removing it reduces Tromsø precision and F1 in the reported ablation, establishing component sensitivity. Because no valley-specific prediction artifact or matched generic regularizer is reported, the experiment does not by itself localize the gain to valley-floor suppression or establish unchanged sensitivity within true avalanche paths.

### 5.2 Cross-System Transferability Across Global Mountain Regimes

A primary challenge in machine learning for natural hazards is model overfitting to localized geographic and environmental training conditions. The AvalCD benchmark spans four distinct mountain systems characterized by starkly contrasting climatic, snowpack, and geomorphic regimes:

- 1. Temperate Alpine Regime (European Alps):** High recreation density, moderate elevation relief, and mixed maritime-continental snowpacks characterized by distinct layer metamorphism and frequent spring wet-snow transitions.

2. **High-Altitude Continental Regime (Pamir Mountains, Central Asia):** Extreme vertical relief ( $> 2,000$  m descent paths), hyper-arid winter climate, deeply faceted depth hoar, and massive powder/slab avalanche cycles capable of burying entire valleys.
3. **Maritime Sub-Arctic Regime (Tromsø, Norway):** Coastal mountains with heavy, wind-scoured maritime snowpacks, frequent rain-on-snow events, and polar darkness throughout mid-winter.
4. **Polar Fjord Alpine Regime (Nuuk, Greenland):** Sub-polar maritime conditions with persistent high-velocity winds and coastal fjord relief.

When evaluated zero-shot on the unseen Pamir Mountains and Scandinavian Arctic, TopoRadar-Net maintains robust segmentation fidelity and high polygon completeness. This cross-system resilience suggests that conditioning on fundamental physical coordinates ( $[\cos \psi, \theta, \sin \alpha, \cos \alpha, \theta_{\text{align}}]$ ) provides consistent topographic coordinates. Because the physical laws governing radar wave reflection and terrain slope apply across mountain systems, TopoRadar-Net generalizes well from European training data to Central Asian and Arctic mountain environments without requiring site-specific re-tuning or retraining.

### 5.3 Operational Implications for Alpine Civil Protection and Search-and-Rescue

For alpine emergency managers, highway maintenance authorities, and search-and-rescue teams, satellite-based avalanche detection is only valuable if it provides reliable, actionable intelligence during crisis cycles (Eckerstorfer et al. 2019; European Avalanche Warning Services 2022; Techel et al. 2018). Optical remote sensing is consistently thwarted during major blizzard events when rapid avalanche detection is most urgently needed. Sentinel-1 C-band SAR provides the only freely accessible, all-weather observation platform capable of continuous winter surveillance.

In operational disaster response, detection accuracy across different avalanche magnitude classes is of paramount importance. Small sluffs (EAWS Class D1) pose localized risks to backcountry recreationists, whereas large and very large avalanches (Classes D3 and D4) destroy buildings, block transportation arteries, and bury infrastructure. TopoRadar-Net exhibits high instance-level detection hit rates on Class D3 and D4 avalanches, ensuring that major life-threatening events are identified for civil protection authorities.

Table 8 summarizes the quantitative clutter footprint across mountain regimes and outlines an illustrative conceptual triage framework:

1. **Quantified Clutter Suppression in Physical Units:** In operational deployments, false alarms waste critical emergency response resources. Across the evaluated high-relief Pamir Mountains scene ( $359.10 \text{ km}^2$  valid mask), TopoRadar-Net reduces total false-alarm area to  $169.89 \text{ ha}$  ( $1.70 \text{ km}^2$ ), achieving a substantial  $72.88 \text{ ha}$  (30.0%) reduction in false-alarm footprint compared to the Attention U-Net baseline ( $242.77 \text{ ha}$ ) and a  $132.74 \text{ ha}$  (43.9%) reduction compared to SiamUNet-conc ( $302.63 \text{ ha}$ ) on the nominal 10 m grid

(corresponding to  $55.63 \text{ ha}$  and  $101.33 \text{ ha}$  reductions under true affine pixel areas). In Tromsø ( $245.87 \text{ km}^2$  valid mask), TopoRadar-Net detects  $270.09 \text{ ha}$  of true avalanche debris (capturing 95.8% of Class D4 events), maintaining a low false-alarm density of  $0.36 \text{ ha per km}^2$  of rugged terrain. Evaluated on the raw GeoTIFF affine resolutions ( $85.15 \text{ m}^2/\text{px}$  in Tromsø,  $76.34 \text{ m}^2/\text{px}$  in Pamir), physical false-alarm areas are  $74.77 \text{ ha}$  and  $129.69 \text{ ha}$  respectively, yielding identical false-alarm rates ( $0.36$  and  $0.47 \text{ ha/km}^2$ ) because pixel counts scale identically in numerator and denominator. We openly disclose that in Tromsø, TopoRadar-Net’s false-alarm footprint ( $87.82 \text{ ha}$ ) is higher than Attention U-Net ( $68.34 \text{ ha}$ ) and ResU-Net ( $65.17 \text{ ha}$ ), reflecting an operational trade-off where TopoRadar-Net yields high detected true debris area ( $270.09 \text{ ha}$ ) and maintains 95.8% Class D4 completeness (tying ResU-Net at 95.8%, behind Attention U-Net’s 100.0%), incurring modest additional perimeter clutter in complex fjord coastal transitions.

2. **Conceptual Decision-Support Triage Framework:** Table 8 outlines an illustrative conceptual triage framework for decision support: Tier 1 ( $p < 0.40$ ) corresponds to low-confidence background detections; Tier 2 ( $0.40 \leq p < 0.70$ ) highlights moderate-probability regions warranting expert review or optical cross-examination; and Tier 3 ( $p \geq 0.70$ ) indicates high-probability major debris candidates along key infrastructure routes (such as the E8 Arctic Highway in Norway or the M41 Pamir Highway in Tajikistan). We clarify that this represents an unvalidated conceptual framework rather than an automated dispatch protocol. Across the verified Tromsø benchmark, detected Class D4 avalanche events exhibit 95.8% detection completeness across training seeds, confirming high model sensitivity to major release events.
3. **Model-Forward Inference Throughput Profile:** TopoRadar-Net maintains a compact footprint of  $3.46\text{M}$  parameters ( $3,455,729$  parameters). Profiling shows a full-scene sliding-window model inference time of  $1.80 \pm 0.02$  seconds ( $668.5$  patches/s) on Apple Silicon MPS hardware (artifact `inference_profile.json`), confirming that core neural network computation imposes negligible overhead within regional processing pipelines.

### 5.4 Limitations and Future Research

While TopoRadar-Net demonstrates substantial advances in alpine SAR change detection, several physical and technical limitations warrant disclosure:

- **Augmentation Convention:** The released checkpoints use the disclosed checkpoint-era convention in which aspect sine/cosine components are sign-adjusted under reflections while the precomputed orbital look-alignment channel is spatially transformed without directional recomputation; orthogonal rotations similarly preserve channel values. The resulting training tensor is not fully equivariant under these transformations. Consequently, the directional results are interpreted as empirical sensitivity rather than a causal validation of the manifold representation. A future

**Table 8** Conceptual decision-support triage framework (illustrative design, unvalidated for automated operational dispatch) and quantitative false-alarm clutter footprint across regional mountain regimes in physical land-management units (hectares and km<sup>2</sup>), computed on the evaluated Sentinel-1 valid scene footprint (Tromsø 245.87 km<sup>2</sup>, Pamir 359.10 km<sup>2</sup> on nominal 10 m grid; affine-scaled areas scale proportionally with identical ha/km<sup>2</sup> rates).

| Triage Level                                                                                                                     | Score ( $p$ )                                  | Debris Footprint                               | Illustrative Hazard Tier                                                    | Target Stakeholder                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Tier 1</b>                                                                                                                    | $p < 0.40$                                     | $< 0.1$ ha<br>( $< 10$ px)                     | Low confidence; routine background monitoring                               | Warning Services (SLF, NVE)                                                        |
| <b>Tier 2</b>                                                                                                                    | $0.40 \leq p < 0.70$                           | $0.1\text{--}1.0$ ha<br>( $10\text{--}100$ px) | Moderate confidence; recommended expert review or optical cueing            | Regional Forecasters                                                               |
| <b>Tier 3</b>                                                                                                                    | $p \geq 0.70$                                  | $\geq 1.0$ ha (Major debris)                   | High confidence; prioritized inspection along transport corridors           | Civil Protection / Road Auth.                                                      |
| <b>Regional False-Alarm Footprint in Evaluated Scene (Valid Area: Tromsø 245.87 km<sup>2</sup>, Pamir 359.10 km<sup>2</sup>)</b> |                                                |                                                |                                                                             |                                                                                    |
| Region                                                                                                                           | Method                                         | Detected Debris (ha)                           | False-Alarm Area (ha / km <sup>2</sup> )                                    | False-Alarm Rate (ha / km <sup>2</sup> )                                           |
| Scandinavian Arctic (Tromsø)                                                                                                     | SiamUNet-diff                                  | 218.33 ha                                      | 43.13 ha (0.43 km <sup>2</sup> )                                            | 0.18 ha/km <sup>2</sup> (Misses 121.39 ha)                                         |
|                                                                                                                                  | Attention U-Net<br><b>TopoRadar-Net (Ours)</b> | 261.49 ha<br><b>270.09 ha</b>                  | 68.34 ha (0.68 km <sup>2</sup> )<br><b>87.82 ha (0.88 km<sup>2</sup>)</b>   | 0.28 ha/km <sup>2</sup><br><b>0.36 ha/km<sup>2</sup></b> (Detects 95.8% D4)        |
| Pamir Mountains (Pish)                                                                                                           | SiamUNet-conc                                  | 163.76 ha                                      | 302.63 ha (3.03 km <sup>2</sup> )                                           | 0.84 ha/km <sup>2</sup> (Severe Clutter)                                           |
|                                                                                                                                  | Attention U-Net<br><b>TopoRadar-Net (Ours)</b> | 150.43 ha<br><b>130.25 ha</b>                  | 242.77 ha (2.43 km <sup>2</sup> )<br><b>169.89 ha (1.70 km<sup>2</sup>)</b> | 0.68 ha/km <sup>2</sup><br><b>0.47 ha/km<sup>2</sup></b> (30.0% Clutter Reduction) |

confirmatory rerun should compare a fully equivariant augmentation against the released convention.

- **DEM Quality and Geometric Resolution:** TopoRadar-Net relies on the Copernicus GLO-30 DEM. While GLO-30 provides global coverage, localized vertical errors or smoothing over sharp knife-edge arêtes can propagate inaccuracies into the local incidence angle calculation. In regions with available high-resolution airborne LiDAR or stereo-photogrammetric DEMs (1–5 m), topographic conditioning can be further refined.
- **Dielectric Attenuation in Wet Snow Cycles:** Liquid water content (LWC) in snow drastically alters microwave permittivity. During spring rain-on-snow events or warm spells, wet snow absorbs C-band radar waves, reducing backscatter by up to 10 dB across entire mountain faces. While bitemporal differencing partially captures relative roughness, extensive liquid water can mask debris deposits or create wide-area absorption anomalies. Future work should integrate multi-temporal radiometric time-series or polarimetric decomposition parameters ( $H/\alpha$ ) to decouple wet snow transitions.
- **Temporal Revisit Constraints:** Sentinel-1 operates on a 6- to 12-day repeat pass. In rapidly evolving storm cycles, multiple avalanches may release along the same path over several days, aggregating into a single composite deposit. The upcoming launch of Sentinel-1C and Sentinel-1D, alongside the dual-frequency L/S-band NISAR mission, will provide dense sub-weekly revisit cycles, enabling near-continuous temporal monitoring.

## 6 Conclusion

Automated mapping of snow avalanche debris in steep alpine terrain is a vital capability for mountain natural hazard risk reduction and emergency response. In this study, we addressed the long-standing challenge of severe topographic-radar distortion in spaceborne Sentinel-1 C-band SAR change detection by proposing **TopoRadar-Net**, a physics-guided deep change

detection network incorporating multi-scale Radar-Topographic Cross-Attention Blocks (RTCAB) and Geomorphically Bounded Loss (Geo-Loss).

Our key scientific and methodological findings include:

1. **Theoretical Formulation of 2D Convolutional Inadequacy in Mountain SAR:** We formulated (Proposition 2) the geometric limitations of shift-invariant 2D spatial linear convolutions across sloped mountain facets, showing that projecting terrain into a continuous 4D aspect-look manifold ( $\mathcal{M}$ ) provides continuous coordinates to condition feature representations without angular branch-cuts.
2. **Physics-Guided Cross-Attention Suppresses Foreslope Distortion:** By explicitly formulating a continuous directional aspect-look alignment coordinate ( $\theta_{\text{align}}$ ) and local incidence angle ( $\psi$ ), TopoRadar-Net dynamically modulates SAR backscatter change representations, effectively suppressing foreslope layover clutter while amplifying backslope debris surface roughness contrast in steep coastal mountain relief.
3. **Geomorphic Regularization Suppresses Valley False Alarms:** The integration of a soft geomorphic boundary penalty regularizes the network against physically impossible landforms ( $< 5^\circ$  valley floors and  $> 65^\circ$  vertical cliffs), providing a confirmed +3.16% F1 gain ( $p = 0.002$ ) by suppressing non-hazard false positives.
4. **Cross-System Generalization and Stratification Findings:** Across the AvalCD benchmark spanning four global mountain systems, TopoRadar-Net achieves  $63.79\% \pm 1.88\%$  pooled macro F1, exceeding Attention U-Net (63.41%), SiamUNet-conc (63.13%), ResU-Net (62.92%), and SiamUNet-diff (54.87%). TopoRadar-Net achieves a statistically significant 3-seed pooled gain over SiamUNet-diff (+8.92%,  $p = 0.021$ ), while paired multi-seed testing against competitive baselines confirms cross-system performance parity ( $p = 0.869$  vs Attention U-Net,  $p = 0.459$  vs ResU-Net). Fine-grained topographic stratification characterizes the regional performance patterns, confirming consistent  $> +5\%$  F1 advantages on back-slopes in both mountain systems (+5.85% in Tromsø,



driven by recall; +5.09% in Pamir, driven by precision), while disclosing that extreme high-relief continental shadow (Pamir > 45°) dampens auxiliary feature utility, corroborating benchmark findings.

5. **Operational EAWS Decision Matrix and Real-World Clutter Suppression:** We formulated a 3-tier operational triage framework in physical units (hectares), demonstrating a 30.0% (72.9 ha nominal grid, 55.6 ha affine physical reduction) false-alarm reduction in the high-altitude Pamirs and high instance completeness on catastrophic Class D4 avalanches (100% detection in 2 of 3 seeds; mean 95.8%), operating with a lightweight 3.46M parameter footprint and 1.80 s full-scene inference latency.

By bridging physical radar geomorphology with efficient deep learning architectures, TopoRadar-Net provides a dependable, all-weather, day-and-night remote sensing solution for monitoring dangerous snow avalanches across mountain regions worldwide.

## Declarations

**Ethical Approval and Consent:** Not applicable.  
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**Availability of Data and Materials:** The AvalCD benchmark dataset is publicly deposited on Zenodo under DOI: <https://doi.org/10.5281/zenodo.15863589>. Sentinel-1 Synthetic Aperture Radar data and Copernicus GLO-30 Digital Elevation Models are openly provided under the Copernicus Open Access Policy.

**Code Availability:** All neural network implementations, baseline architectures, training protocols, and verification suites supporting this study are openly maintained in the project repository at: <https://github.com/akssha74/TopoRadar-Net-AvalCD>. Pretrained model weights for all three independent training seeds and primary verification artifacts are preserved alongside the repository.

**Authors' Contributions:** Akshay Sharma: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. Lalji Prasad: Supervision, Methodology, Resources, Validation, Writing – Review & Editing. All authors read and approved the final manuscript.

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